

OPPS Concept

Encapsulation in Java :-

- Encapsulation is the process of wrapping variables and methods in a single unit (class).
- It is used for data hiding by declaring variables as private.
- Access to variables is provided using public getter and setter methods.
- It increases security, flexibility, and reusability of code.
- Encapsulation ensures that implementation details are hidden from the user.

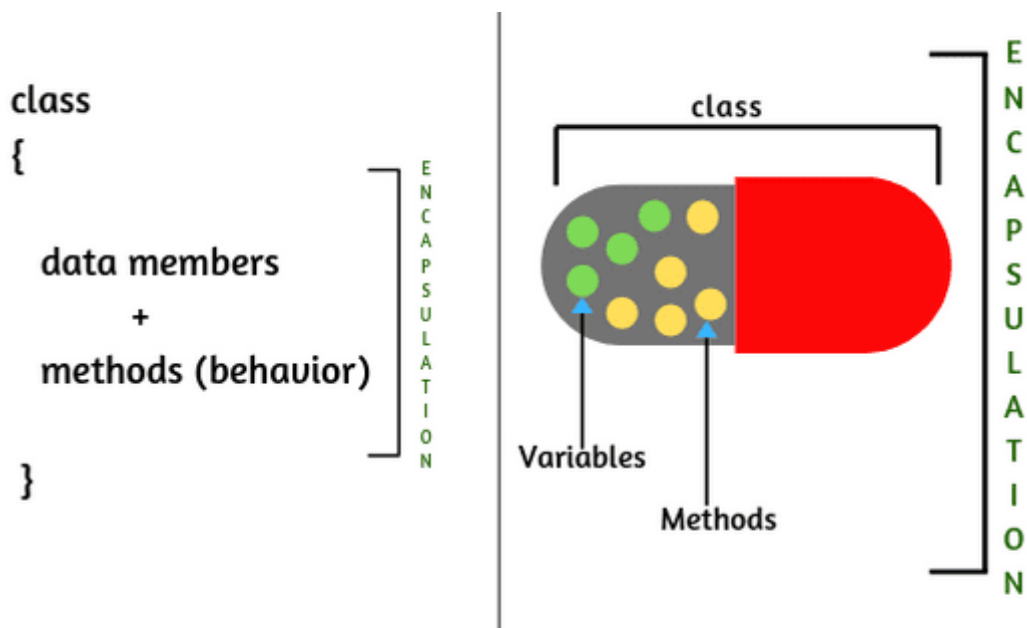
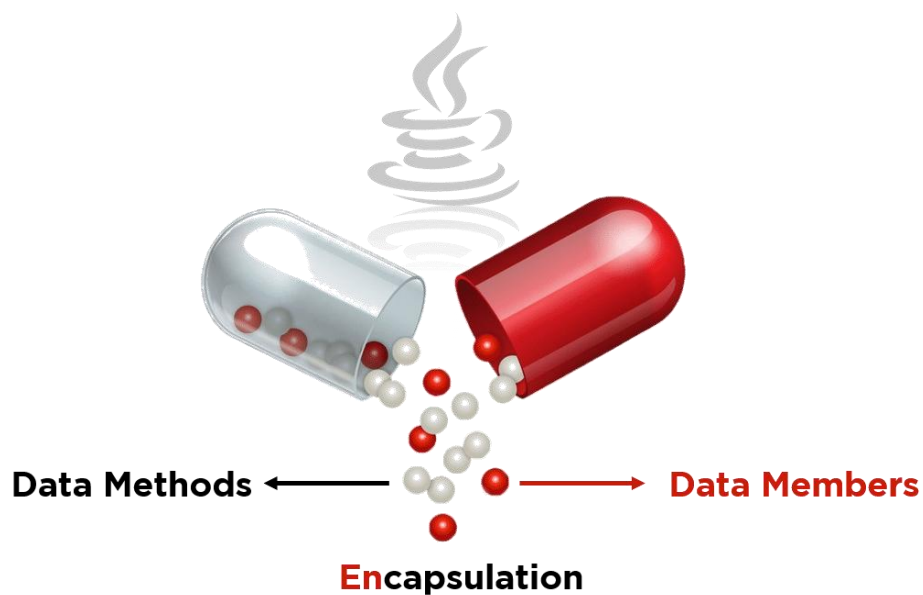


Fig: Encapsulation

Syntax –

```
<Access_Modifier> class <Class_Name> {  
    private <Data_Members>;  
    private <Data_Methods>;  
}
```

Advantages of Encapsulation :-

- **Data Hiding:** Keeps internal data safe from unauthorized access.
- **Controlled Access:** Access to variables is possible only through getter and setter methods.
- **Improved Security:** Sensitive data cannot be modified directly.
- **Increased Flexibility:** Implementation of methods can be changed without affecting outside code.
- **Better Maintainability:** Code becomes easier to maintain and modify.
- **Reusability:** Encapsulated classes can be reused in different programs without changing internal logic.

Disadvantages of Encapsulation :-

- **More Code:** Requires writing additional getter and setter methods.
- **Slightly Slower:** Accessing private variables through methods is a bit slower than direct access.
- **Complexity:** For small programs, using encapsulation may add unnecessary complexity.
- **Memory Overhead:** More objects and methods can slightly increase memory usage.

Example –

```
package oops;

public class encapsulation {
    private long stu_aadhar_no; //instance variable/ global variable
    private String stu_account_number;

    void set_addharno(long stu_aadhar_no)    //setter method
    {
        this.stu_aadhar_no= stu_aadhar_no;
    }

    void set_account_number(String stu_account_number)
    {
        this.stu_account_number=stu_account_number;
    }

    long get_addharno() //getter method
    {
        return stu_aadhar_no;
    }

    String get_account_number()
    {
        return stu_account_number;
    }

    public static void main(String[] args) {
        encapsulation obj=new encapsulation();
        obj.set_addharno(4091764583501);
        System.out.println("Student Aadhar Number is: "+obj.get_addharno());

        obj.set_account_number("BOI5486245");
        System.out.println("Student Account Number is: "+obj.get_account_number());
    }
}
```

Output –

```
Student Aadhar Number is: 409176458350
Student Account Number is: BOI5486245
```

Task –

```
package practical;

import java.util.Scanner;

public class encapsulation_task_emp {
    private long employee_id;
    private float employee_salary;

    void set_empid(long employee_id){
        this.employee_id=employee_id;
    }

    long get_empid(){
        return employee_id;
    }

    void set_empsal(float employee_salary) {
        this.employee_salary=employee_salary;
    }

    float get_empsal(){
        return employee_salary;
    }

    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        long id;
        float sal;

        System.out.println("Enter the Employee_Id: ");
        id=sc.nextInt();

        System.out.println("Enter the Employee_Salary: ");
        sal=sc.nextFloat();

        encapsulation_task_emp obj=new encapsulation_task_emp();
        obj.set_empid(id);
        System.out.println("Employee Id is: "+obj.get_empid());
        obj.set_empsal(sal);
        System.out.println("Employee Salary is: "+obj.get_empsal());
    }
}
```

Output –

Enter the Employee_Id:

101

Enter the Employee_Salary:

25000

Employee Id is: 101

Employee Salary is: 25000.0